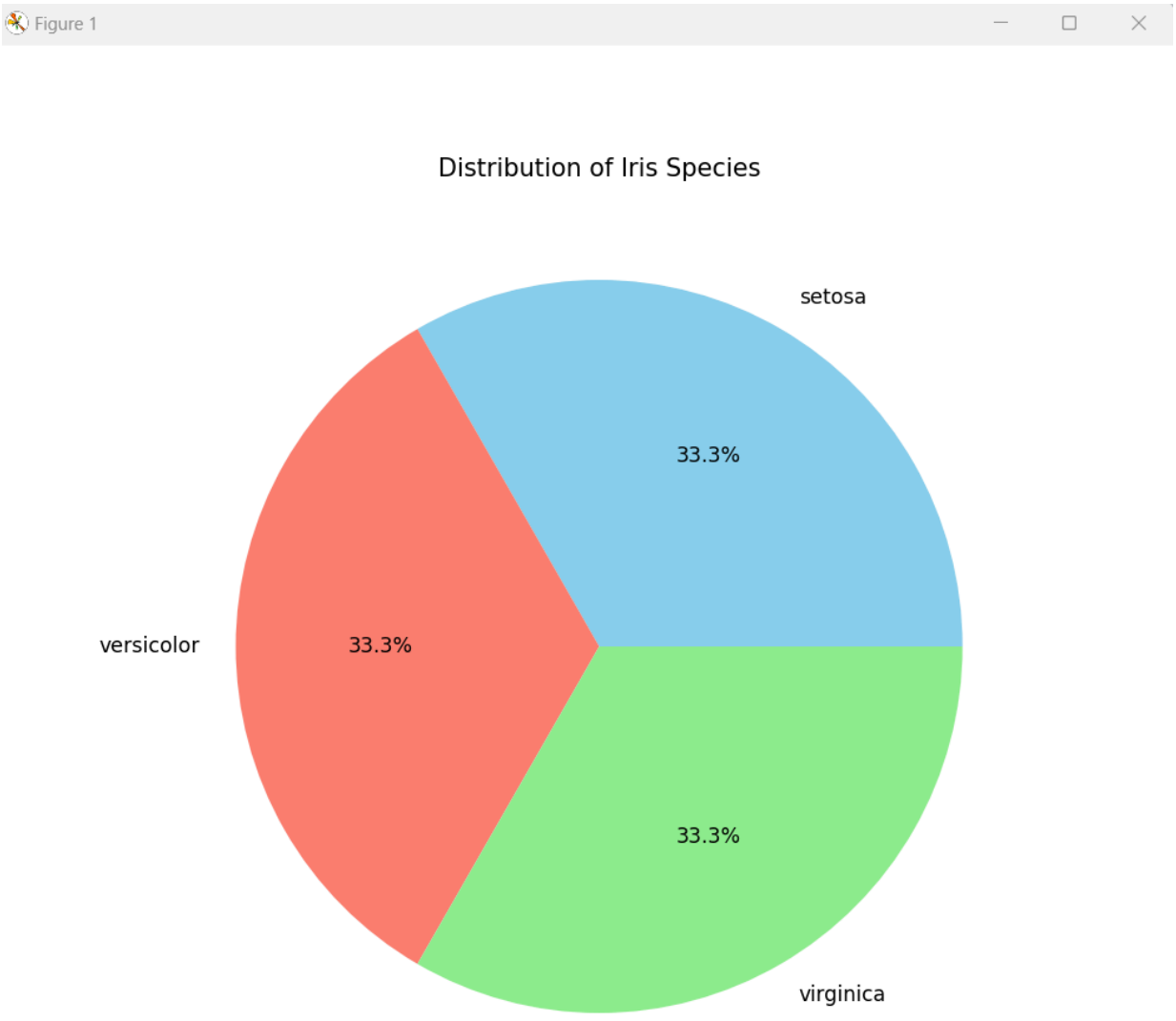
Naive Bayes Algorithm

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score
iris = datasets.load_iris()
iris_df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
print(iris_df.head())
iris_df['species'] = iris.target
species_names = {0: 'setosa', 1: 'versicolor', 2: 'virginica'}
iris_df['species'] = iris_df['species'].map(species_names)
X = iris.data
y = iris.target
print (X.shape)
print (y.shape)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=9)
nv = GaussianNB()
nv.fit(X_train, y_train)
print("GaussianNB classifier:", nv)
y_pred = nv.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy of GaussianNB classifier: {accuracy:.2f}")
species_counts = iris_df['species'].value_counts()
plt.figure(figsize=(8, 8))
plt.pie(
    species_counts,
    labels=species_counts.index,
    autopct='%1.1f%%',
    colors=['skyblue', 'salmon', 'lightgreen'],
    startangle=140
)
plt.title('Distribution of Iris Species')
plt.show()
```

OUTPUT :

```
====== RESTART: C:/Users/Shruti Santosh Patil/Documents/DWM/Naive bayes.py =====
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
(150, 4)
(150,)
GaussianNB classifier: GaussianNB()
Accuracy of GaussianNB classifier: 1.0
```



Naive Bayes Algorithm

```
import pandas as pd
import numpy as np
from sklearn import datasets # type: ignore
iris = datasets.load_iris()
iris_df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
print(iris_df.head())
```

OUTPUT :

```
====== RESTART: C:/Users/Shruti Santosh Patil/Documents/DWM/Naive bayes.py =====
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
```

```
X=iris.data
y=iris.target
print (X.shape)
print (y.shape)
```

OUTPUT :

```
(150, 4)
(150,)
```

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=9) #Split the
dataset
from sklearn.naive_bayes import GaussianNB
nv = GaussianNB()
nv.fit(X_train,y_train)
print("GaussianNB classifier:", nv)
```

OUTPUT :

```
GaussianNB classifier: GaussianNB()
```

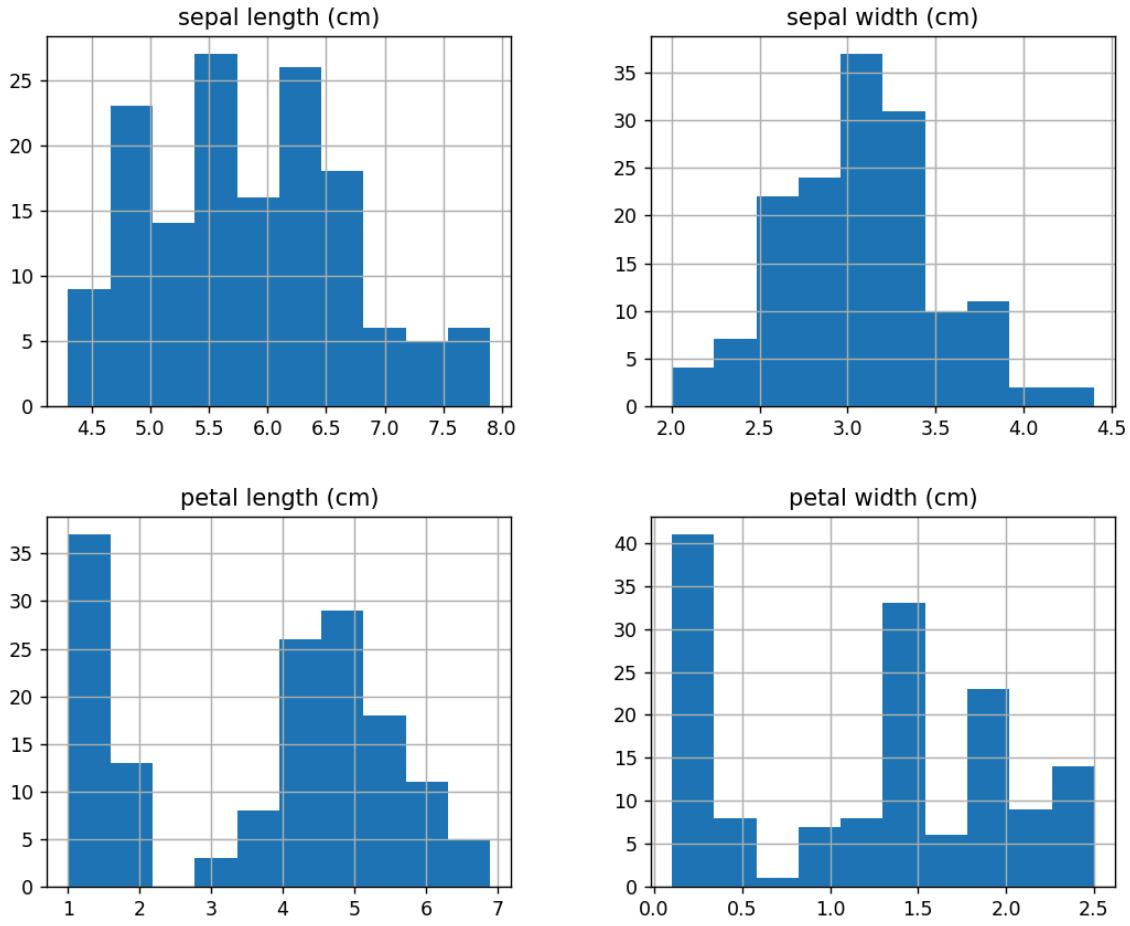
```
from sklearn.metrics import accuracy_score
y_pred = nv.predict(X_test)
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy of GaussianNB classifier:", accuracy)
```

OUTPUT:

```
Accuracy of GaussianNB classifier: 1.0
```

```
import matplotlib.pyplot as plt
iris_df.hist(figsize=(10, 8))
plt.show()
```

OUTPUT:



Naive Bayes Algorithm

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets

iris = datasets.load_iris()
iris_df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
print(iris_df.head())
iris_df['species'] = iris.target
```

OUTPUT :

```
====== RESTART: C:/Users/Shruti Santosh Patil/Documents/DWM/Naive bayes.py =====
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
```

```
X = iris.data
y = iris.target
print (X.shape)
print (y.shape)
```

OUTPUT :

```
(150, 4)
(150,)
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=9)
from sklearn.naive_bayes import GaussianNB
nv = GaussianNB()
nv.fit(X_train, y_train)
print("GaussianNB classifier:", nv)
```

OUTPUT :

```
GaussianNB classifier: GaussianNB()
```

```
from sklearn.metrics import accuracy_score
y_pred = nv.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy of GaussianNB classifier: {accuracy:.2f}")
```

OUTPUT :

```
Accuracy of GaussianNB classifier: 1.0
```

```
import matplotlib.pyplot as plt
species_names = {0: 'setosa', 1: 'versicolor', 2: 'virginica'}
iris_df['species'] = iris_df['species'].map(species_names)
species_counts = iris_df['species'].value_counts()
plt.figure(figsize=(8, 8))
plt.pie(
    species_counts,
    labels=species_counts.index,
    autopct='%1.1f%%',
    colors=['skyblue', 'salmon', 'lightgreen'],
    startangle=140
)
plt.title('Distribution of Iris Species')
plt.show()
```

OUTPUT :

